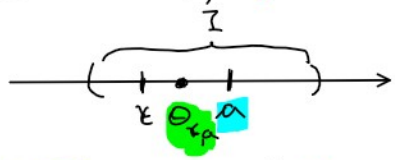
Lab 1 - notițe

Formula lui Taylor

I = interval deschis; $C^k(I) = \{ f : I \to \mathbb{R} : \exists f^{(i)}, i = \overline{0,k};\ f^{(k)} \text{ cont.} \}$

$\boxed{T_1}$ (formula lui Taylor cu restul Lagrange)

$$f \in C^{n+1}(I),\ a \in I \implies f(x) = \underbrace{T_n f(x)}_{\substack{\text{pol. Taylor de ord. } n \\ \text{în jurul lui } a}} + \underbrace{R_n f(x)}_{\text{rest}},\ x \in I, \text{ unde}$$



$$T_n f(x) = f(a) + \frac{f'(a)}{1!}\cdot(x-a) + \dots + \frac{f^{(n)}(a)}{n!}\cdot(x-a)^n$$

$$R_n f(x) = \frac{f^{(n+1)}(\theta_{x,a})}{(n+1)!}\cdot(x-a)^{n+1},$$ $\theta_{x,a}$ = pct. intermediar este între x și a

$T_n f(x)$ = taylor(f, x, a, 'order', n+1) → apel în Matlab/Octave

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>> help @sym/taylor
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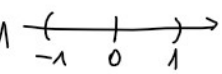
Serii de puteri MacLaurin ($a=0$)

$$e^x = 1 + \frac{x^1}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots,\quad x \in \mathbb{R}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^n}{(2n+1)!}\cdot x^{2n+1} + \dots,\quad x \in \mathbb{R}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^n}{(2n)!}\cdot x^{2n} + \dots,\quad x \in \mathbb{R}$$

$$\underbrace{\ln(1+x)}_{\log} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{(-1)^{n-1}}{n}\cdot x^n + \frac{(-1)^n}{n+1}\cdot x^{n+1} + \dots,\quad |x|<1$$



$$\frac{1}{1+x} = 1 - x + x^2 - \dots + (-1)^n\cdot x^n + \dots,\quad |x|<1$$

$$\frac{1}{1-x} = 1 + x + x^2 + \dots + x^n + \dots,\quad |x|<1$$

$$(1+x)^k = 1 + \sum_{n=0}^{\infty} \underbrace{\binom{k}{n}}_{\frac{k\cdot(k-1)\cdot\ldots\cdot(k-(n-1))}{n!}}\cdot x^n,\quad |x|<1$$

$(k \in \mathbb{R})$

ex: $(1+x)^{-\frac{1}{2}} = \dfrac{1}{\sqrt{1+x}}$

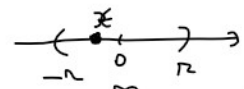
T2 (crit. lui Leibniz pt. serii alternante)

$$S = \underbrace{\sum_{k=0}^{\infty} (-1)^k a_k}_{\text{conv.}}, \quad \text{unde } (a_k)_{k\in\mathbb{N}} \text{ desc. cu } \lim = 0 \quad (a_k \searrow 0,\ k \to \infty)$$

$$S_n = \sum_{k=0}^{n} (-1)^k a_k \Rightarrow |S - S_n| \le a_{n+1}.$$

T3 (teor. lui Abel pt. serii)

$$f(x) = \sum_{k=0}^{\infty} a_k \cdot x^k, \quad |x| < r$$



$$\sum_{k=0}^{\infty} a_k \cdot r^k \text{ conv.} \Longrightarrow \lim_{x \nearrow r} f(x) = \sum_{k=0}^{\infty} a_k \cdot r^k.$$